Product Quality 2026: What the Data Shows and a Quality Model

Management update v2 | QA Architecture | 25 October 2026

Bugs got more severe; customers still find them first

+49%

P1/P2 customer-found bugs

Jan-Sep 2026 vs Jan-Sep 2025

+13%

all customer-found bugs

same window; MI investigations removed

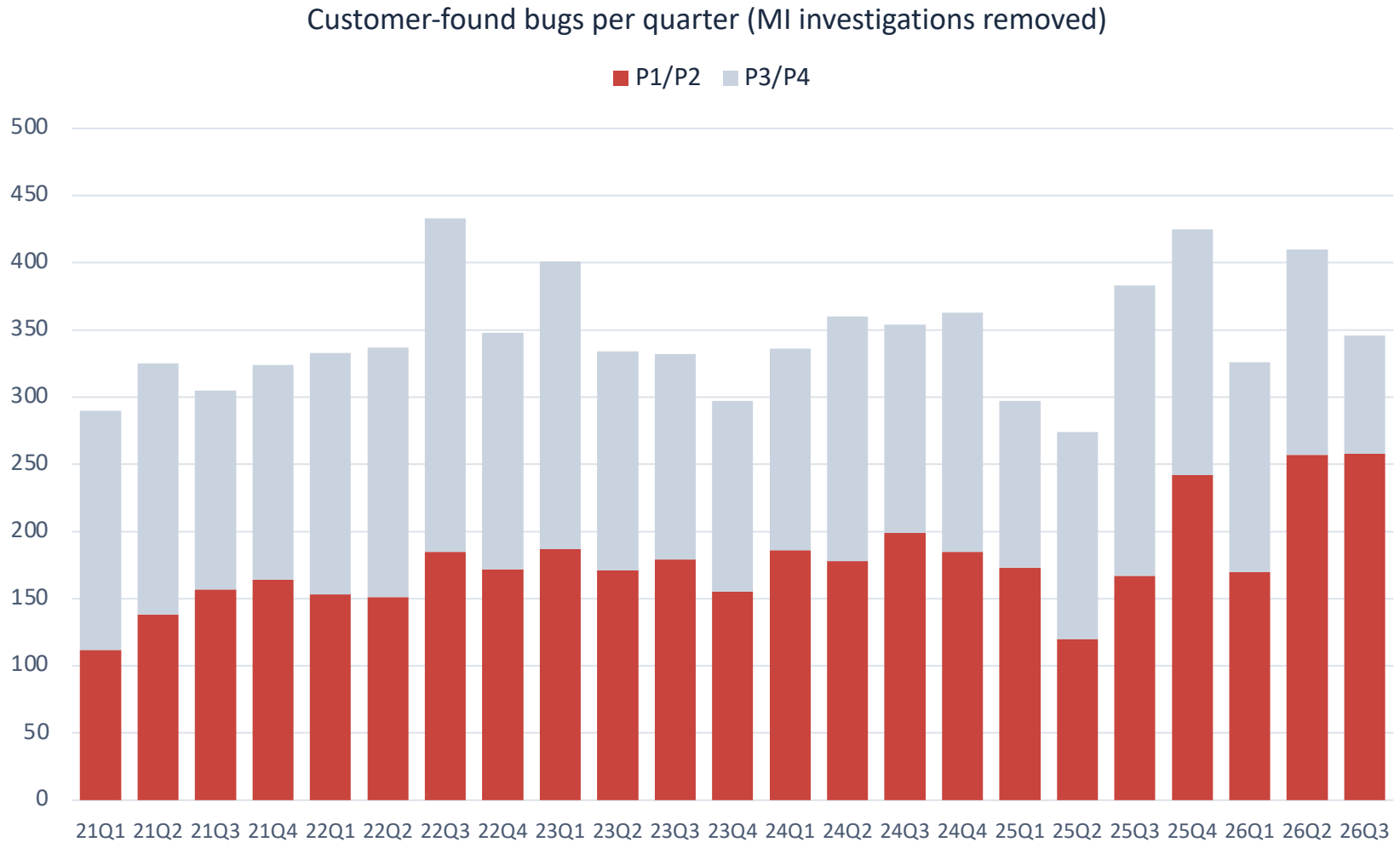
96%

of P1 incidents reach us via customer cases

ServiceNow, since 2025

1. **Severity, not volume:** P1/P2 rose sharply; total volume rose a little and fell in Q3.
2. **Not AI per change:** matched comparisons show no higher bug rate for AI-made changes.
3. **Proposal:** a quality model with a common baseline, risk-based rules, team adaptation, and 5 experiments.

Severity broke the trend in 2026; volume did not



258 vs 167

P1/P2 in Q3 2026 vs Q3 2025

346 vs 383

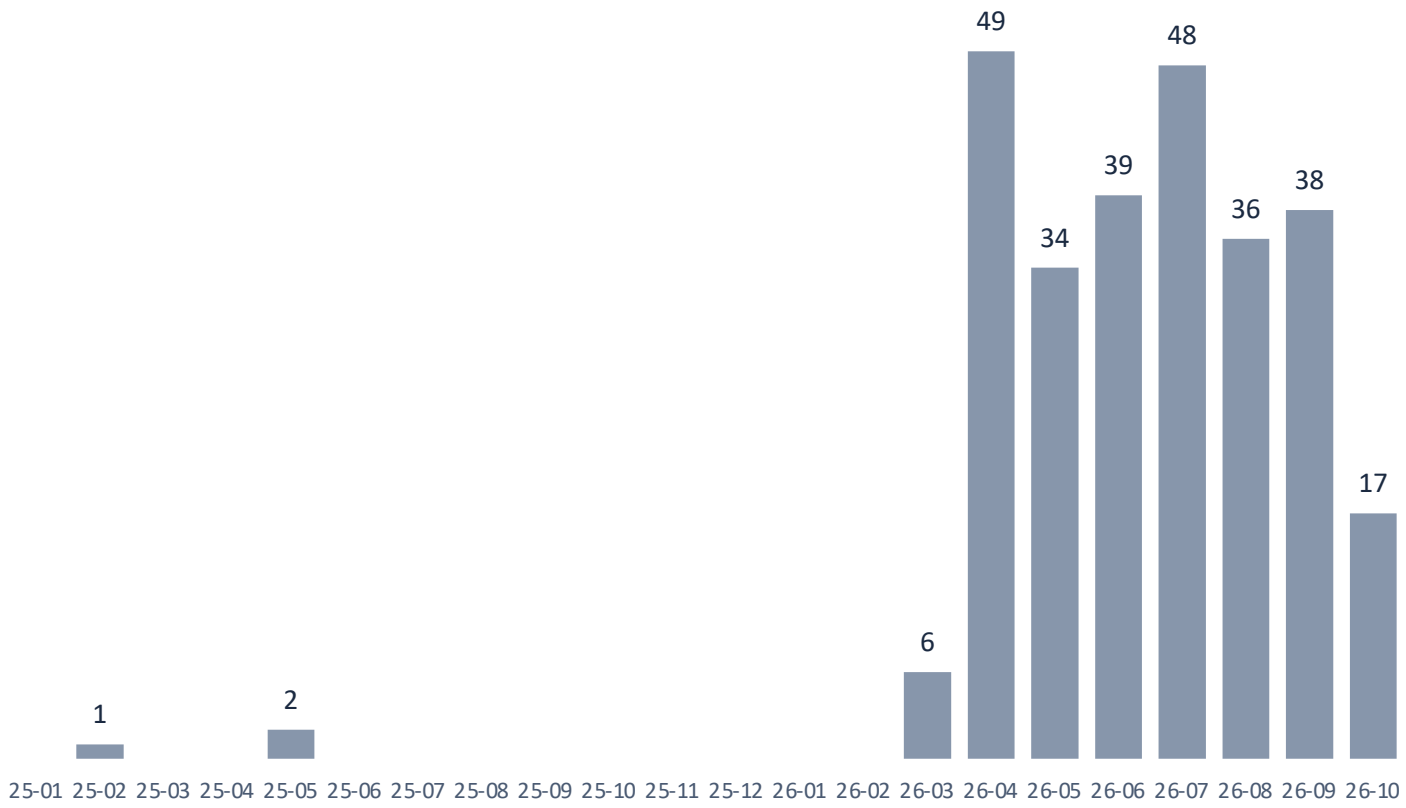
all bugs in Q3 2026 vs Q3 2025

75%

P1/P2 share in Q3 2026 (about 50% in 2023-24)

Correction: our first headline counted MI investigations as bugs

MI investigation tickets logged as customer bugs, per month



Earlier (H1)

+51% bugs, +72% P1/P2

Corrected (Jan-Sep)

+13% bugs, +49% P1/P2

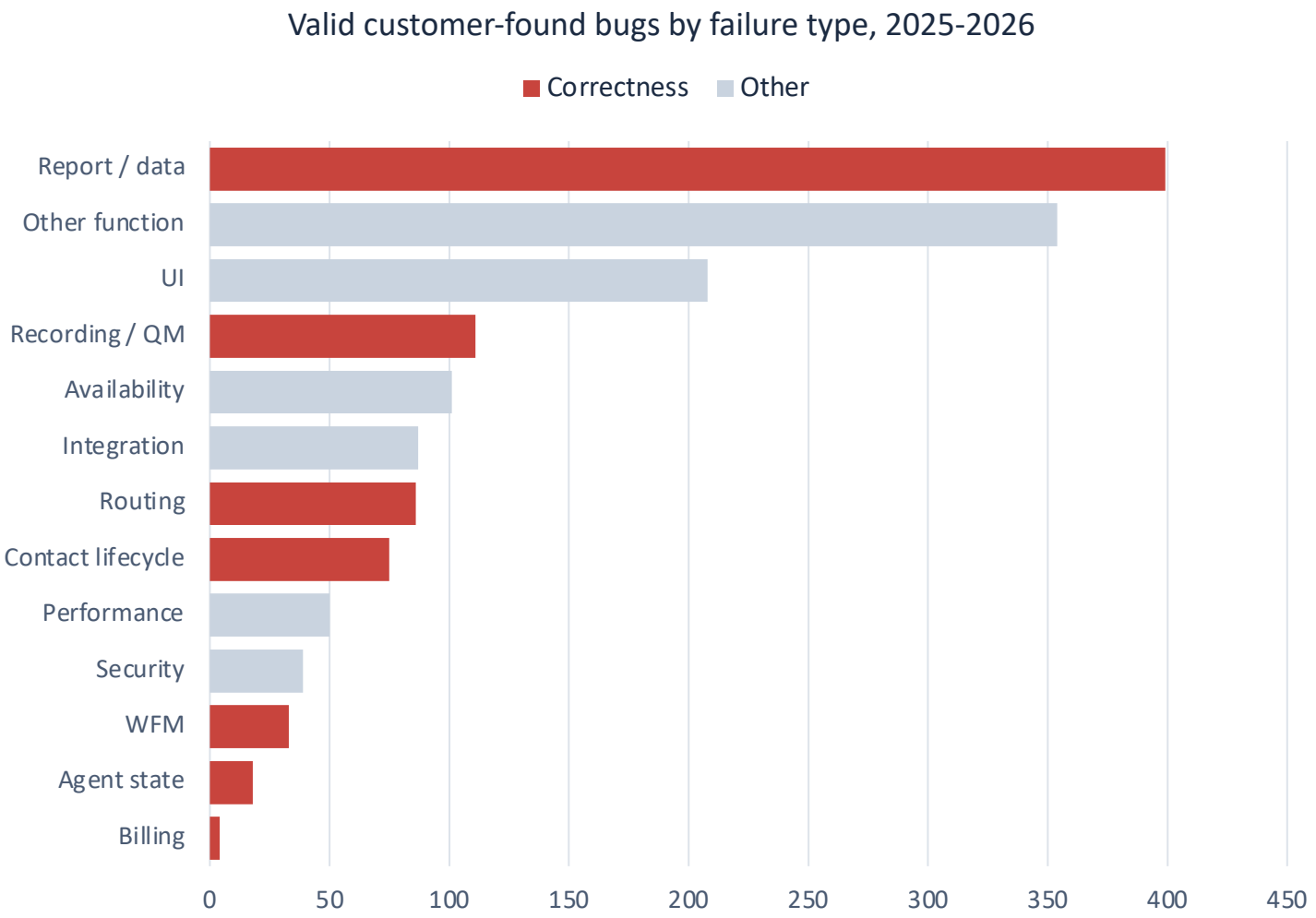
Why

From March 2026, every major-incident investigation is logged as a bug with a customer case

Lesson

We now audit what enters a count before we trend it

Almost half of valid bugs are business-correctness failures



73%

of routing bugs are P1/P2 (highest)

11

incidents per agent-state bug

52% -> 70%

P1/P2 share of correctness bugs, 2025 -> 2026

AI Quality Discovery | Data as of 2026-10-25 | Sources: Jira, Confluence, GitHub, ServiceNow

Teams differ too much for one workflow

1.1 - 10.4

incidents per bug

10 - 93%

P1/P2 share

6 - 79%

invalid bug share

3 - 31

days to fix (median)

Range across 40 teams. The VP explains only 17-46% of this variation; most of it is inside a VP.

High impact

9 teams

Severity and incidents above median

Fast growing

8 teams

Valid bugs +50% or more; part is re-homing

High noise

5 teams

Half or more of bugs invalid

Stable

18 teams

No extra signal

AI: what is fact, what is correlation, what is hypothesis

FACT	AI use grew from about 0% to 31-47% of work in 2026
FACT	AI labels on bugs mark AI-made fixes, not AI-caused bugs
NOT SUPPORTED	AI changes cause more bugs per change (matched ratio 0.78-0.84)
NOT SUPPORTED	AI adoption caused the 2026 rise (adopting teams stayed flat)
WEAK SIGNAL	Same author: AI PRs need more follow-up fixes (1.45, CI touches 1)
MECHANISM	AI PRs are 3.7x larger and 47% of PRs into hotspot code; review does not grow with size

Quality = 8 outcome dimensions, each tied to a decision

Dimension	Decision it supports	2025 -> 2026
Business correctness	Where to invest first	295 -> 325 valid; P1/P2 52% -> 70%
Availability	Reliability, release risk	Counts only (access)
Performance	Capacity work	No SLO data yet
Detection	Monitoring investment	96% of P1 via customer cases
Recovery	Response capacity	P1 71% < 4 h; fix median 10 days
Change safety	Review depth for fixes	1 in 4 causes is a fix; emergency x2.9
Non-recurrence	Do fixes hold	1-2% return; flat
Expectation fit	Requirements and docs	~15% of reports are real gaps

Code quality, coverage and tool scores are inputs that may explain these outcomes. They are not quality.

Common where it adds value, different where risk differs

Common baseline

All teams

- One intake rule: defect vs MI vs expectation gap
- Each bug: type, detected-by, cause, customers
- RCA for P1/P2 (already uniform)
- Team -> service -> code map

Risk-based layer

Any team, by risk

- High-severity correctness failures
- Hotspot code: review scales with size
- Bug fixes: regression evidence
- Emergency changes: follow-up review

Team adaptation

By archetype

- High impact: detection and recovery
- Fast growing: check re-homing, then triage
- High noise: expectation gaps first
- Stable: baseline only

Teams choose how to meet a rule. The model fixes what must be true and how it is measured. Rules apply to all authors, human or AI.

Five small experiments before any rollout

1	**Intake rule and required fields**	3 teams, one per archetype	Bugs with type, detected-by, cause
2	**Monitoring for one correctness type**	1 high-impact team	Share found before customers
3	**Review depth scaled to size in hotspot code**	2 teams vs 2 matched	90-day follow-up fixes per PR
4	**Extra review for bug fixes**	2 teams vs 2 matched	Fix-caused share of new bugs
5	**Expectation-gap triage**	1 high-noise team	Repeat 'works as designed' reports

1 quarter each, with control teams and a stop rule. A practice spreads only when it beats its control.

What we need from management

1. **Approve experiments X1-X5 in volunteer teams**

   1 quarter; no org-wide change yet

2. **Grant ServiceNow record-level read for research**

   Needed for who was affected and change-to-incident links

3. **Name an owner for the team-to-service-to-code map**

   No such map exists today

4. **Name a sponsor and a monthly quality review**

   Each team vs its own baseline, 8 dimensions

Timeline

Now	Start X1-X5
2027-Q1	Re-test the AI effect (90 days)
2027-04	Review experiments vs controls
Then	Spread what worked; model v1